Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Brady, Joseph V. Ph.D.		
Project Title:	Psychosocial Performance Factors in Space Dwelling Groups		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI Dr. Joseph Brady passed away in July 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	03/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hienz, Robert (The Johns Hopkins University School of Medicine) Hursh, Steven (SAIC)		
Grant/Contract No.:	NCC 9-58-NBPF00403		
Performance Goal No.:			
Performance Goal Text:			
	(1) Original Aims of the Project Despite uncertainties regarding the performance requirements of projected space laboratories, work stations, interplanetary vehicles and settlements beyond the Earth's atmosphere, a common feature of spaceflight endeavors over the next half century will be extended stays by human groups in extraterrestrial habitats. The imperatives and opportunities associated with configuring effective psychosocial performance models in support of such space dwelling groups are best served by research approaches that are both heuristic and innovative. The development of functional human and ecological models for such space dwelling groups must, in the first instance, be based upon sound scientific principles with research objectives focused upon the management of semi-permanent as well as permanent groups involved in both operational and space science missions. The research methodology involves development of a distributed interactive multi-person simulation in computer generated environments as an		

Task Description:	experimental test bed for modeling psychosocial performances within and between space-dwelling and Earth- based groups. The simulation approach provides an automated means of setting the context for the analysis of performance in space-dwelling groups and monitoring electronically the effects of varying experimental conditions that alter psychosocial interactions. Distributed interactive simulation experiments characterize the effects of variations in the structure and function of communication channels within and between space-dwelling and Earth-based groups as well as the effects of stressful environmental and behavioral interactions upon psychosocial performance effectiveness. Simulation experiments also determine the effects of variations in the appetitive/aversive characteristics of incentive control systems as well as the effects of selection, training, and experience within and between space-dwelling and Earth-based groups. Communication modes, frequencies, durations, and content are recorded and analyzed with performance effectiveness evaluations based upon assigned group task outcome measures. Conceptual and methodological advances that effectively promote psychosocial and ecological stability will ultimately benefit larger societal units, including those that remain Earth-bound, by enhancing an educational and training technology that assures communication of an expanded generalizeable knowledge base. (2) Key Findings of the Project The results of these studies with scenario tasks requiring identification of geologic samples designated by five different rules in each region, showed clearly that cooperative and productive psychosocial interactions could be maintained between individually isolated and dispersed crewmembers in the simulated task environment. All experimental flight crews actively engaged in communicating and effective problem-solving over extended time intervals without benefit of one another's physical presence. In addition, the investigations of communication modality constraints indicated that, with the scenarios tested, there was a high degree of interchangeability between the available communication modes. For example, the effect of selectively removing text or audio communication was to increase the number of audio messages somewhat when text was removed, and produce a clear increase in text messaging when audio was removed. The overall performance evaluation however, showed no consistent effect upon crew total grade values of eliminating either text or audio. (3) Impact of these Findings The experiments conducted during the past year have extended these initial findings by testing the limits of communication constraints as well as the effects upon performance and psychosocial adaptation of removing all accustomed communication modalities. Although there was a reduction in overall geologic specimen collection and analysis activity under such communication constraints, all crews were able to maintain performance effectiveness at 80% or greater compared to baseline levels despite the "system failure" that made all accustomed modes of communication unavailable. In the absence of the accustomed modes of communication, the individual members of each crew made use of the 'indexical' features of the 'Navigation Network' to maintain effective crew performance levels. Indexical or deictic communication depends upon the representation of objects and/or events in terms of their relationship with an agent with the meaning of the communication embedded in the historical context of relevant concrete actions and outcomes. (4) Proposed Research Plan for the Coming Year The experiments to be conducted during the coming year will extend the distributed interactive simulation studies to analyze the effects of selective communication between crew members and mission control as these are influenced by changes in crew configuration and incentive conditions. The analyses will focus upon both individual crew member performance and psychosocial changes in the patterns of distributed interactive communication.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research conducted within the context of distributed interactive simulation models will provide the basis for developing effective patterns of communication and problem solving strategies as well as a range training procedures to enhance problem solving effectiveness. The Earth-based applications of this research will extend to the small operational group selection and training process as well as the management of stressful interactions and the maintenance of group cohesion and productivity. Not only can the outcome of these studies be expected to have an important impact on safety and the quality of life in many Earth-based applied settings, but larger societal units will ultimately benefit from the resulting conceptual and methodological advances that effectively promote social and ecological stability while concurrently enhancing an education and training technology that assures effective communication of an expanded generalizeable knowledge base.
Task Progress:	The progress of the project's research over the last funded year has focused on experiments conducted to test the limits of communication constraints as well as the effects upon performance and psychosocial adaptation of removing all accustomed communication modalities. Although the results obtained during each flight of their 3-flight missions with all communications removed during the second flight showed that there was a reduction in overall geologic specimen collection and analysis activity under such communication constraints, all crews were able to maintain performance effectiveness at 80% or greater compared to baseline levels despite the "system failure" that made all accustomed modes of communication unavailable during the second flight. The individual members of each of the three crews without benefit of the 'Communication Network' made use of the 'indexical' features of the 'Navigation Network' to maintain effective crew performance levels. Indexical or deictic communication depends upon the representation of objects and/or events in terms of their relationship with an agent and upon the contingent outcome of concrete actions relevant to that relationship. With the Communications Network operating normally, messaging between crewmembers by text, audio and/or whiteboard provided objective descriptions of locations and geologic specimen characteristics without reference to the communicating agents. By contrast, indexical countermeasures in response to Communication Network 'failure' were characterized by representation of objects and events in relation to the agents involved, with the meaning of the communication embedded in the historical context of relevant concrete actions and outcomes. With Crew AG, the indexical countermeasure adopted in response to the Communications Network failure involved the Lander circling repeatedly in the vicinity of a valued geologic specimen until the Rover moved into the area and collected the geologic specimen marked by the Lander leg. Similarly, when the Rover required access to the Lander for refueling or specimen storage, the Rover circled in the vicinity until the Lander opened its loading door for Rover boarding. Crew AM developed a similar array of indexical countermeasures to the communication constraints with Rover and Lander engaging in a series of reciprocal signaling moves. With Crew AQ, indexical 'line-of-sight' procedures adopted by the Rover helped to identify and collect specimens without benefit of audio, text, or 'whiteboard'. Self-report Visual Analogue Scale measures for each individual crewmember during post-mission debriefings provided an assessment of individual adjustment as well as overall crew psychosocial adaptation. There was a general elevation across crews in scale ratings reflecting 'stress', 'anger', 'frustration' and similarly 'negativ
Bibliography Type:	Description: (Last Updated: 11/26/2025)

Articles in Peer-reviewed Journals	Brady JV. "Behavioral health: the propaedeutic requirement." Aviat Space Environ Med. 2005 Jun;76(6 Suppl):B13-24. Review. PMID: 15943190 , Jun-2005
Articles in Peer-reviewed Journals	Brady JV, Hienz RD, Hursh SR, Ragusa LC, Rouse CO, Gasior ED. "Distributed interactive communication in simulated space-dwelling groups." Comput Human Behav. 2004 Mar;20(2):311-40. PMID: 14983895 , Mar-2004
Articles in Peer-reviewed Journals	Hienz RD, Brady JV, Hursh SR, Ragusa LC, Rouse CO, Gasior ED. "Distributed communication and psychosocial performance in simulated space dwelling groups." Acta Astronaut. 2005 May-Jun;56(9-12):937-48. PMID: 15835052 , May-2005
Awards	n/a " Cambridge Center Award for Lifetime Contributions to Behavioral Analysis, October, 2004. " Oct-2004
Awards	" Dews Award for Lifetime Research Contributions, American Society of Pharmacology and Experimental Therapeutics (ASPET), Federated American Societies of Experimental Biology (FASEB), April, 2004 " Jan-2005
Awards	" Hake Award for Applied Behavior Analysis, from American Psychological Association (APA), August, 2004 " Jan-2005
Awards	" Young Award for Space Biomedical Research, National Space Biomedical Research Institute (2005). " Jan-2005
Presentation	Brady, J. V. "Behavior analysis in the space age " Aug-2004
Presentation	Brady, J. V., R. D. Hienz, S. R. Hursh, M. J. Banner, E. D. Gasior, & K. R. Spence "Communication Constraints & Indexical Countermeasures in Simulated Space-Dwelling Groups " Jan-2005
Presentation	Brady, J. V., S. R. Hursh, R. D. Hienz, L. C. Ragusa, C. O. Rous, and E. D. Gasior "Distributed Interactive Communication and Psychosocial Performance in Simulated Space-Dwelling Groups " May-2004